FINAL ASSESSMENT - SQL FOR AI & AUTOMATION DOMAIN

MNC Pattern Assessment - TCS, Infosys, Zoho, Accenture, Cognizant, Flipkart

Duration: 2 Hours	Max Marks: 50 Marks	Answer ANY 5 out of 6 Questions
Domain: AI, ML, AI&DS, CSBS	Date: Tuesday	Each Question: 10 Marks

INSTRUCTIONS TO STUDENTS:

1. Answer **ANY 5 questions** out of 6. If you attempt all 6, first 5 will be evaluated.
2. Write the query with explanation in **separate A4 sheets** provided, then execute and show output.
3. Each question is based on real M NC business scenario for AI & Automation roles.
4. Evaluation is based on logic, not just correct output.
5. No use of mobile phones / internet.

MARK SPLIT-UP - PER QUESTION 10 MARKS

Parameter	Marks	What Evaluator Checks - Detailed Rubric
1. Requirement Understanding & Table Identification	2 M	0.5M - Wrong table chosen / misunderstood requirement 1M - Partial understanding, missed JOIN condition / GROUP BY 2M - Fully correct tables, columns, business logic understood
2. Query Logic & Approach (CORE)	4 M	0M - No logic / blank answer 1M - Attempted but wrong logic 2M - Logic partially correct (e.g., JOIN correct but HAVING missing) 3M - Logic correct but not optimal / not using expected function 4M - Perfect optimized logic with correct JOIN type, GROUP BY, Window Function, CTE
3. Syntax & Execution	2 M	0M - Syntax error, query doesn't run 1M - Runs with warnings / minor syntax issues 2M - Clean execution, proper aliases, keywords, no errors
4. Output Accuracy & Presentation	2 M	0M - Wrong output / no output shown 1M - Partial columns, missing ORDER BY, not handling NULLs 2M - Exact expected output, sorted correctly, handles NULLs, can interpret result for business
TOTAL	10 M	

Grading Scale: 9-10 Excellent - MNC Ready | 7-8 Good | 5-6 Average | <5 Needs Improvement

QUESTION 1: E-Commerce Analytics (Amazon / Flipkart Pattern)

Business Context: Find High Value Customers - Customer Lifetime Value (CLV). Standard question in Flipkart and Amazon SQL interviews to test 3-table JOIN and business aggregation.

Table	Columns / Sample Data
Customers	customer_id, customer_name, city, join_date Sample: (1, Arun, Chennai), (2, Priya, Bangalore)
Orders	order_id, customer_id, product_id, amount, order_date, status Sample: (1001, 1, P1, 65000, 2026-08-01, Delivered)
Products	product_id, product_name, category, price Sample: (P1, Laptop, Electronics, 65000)

Question: Display customers who have placed more than 1 order and their total spending is greater than ■50,000. Also show their city and number of orders. Sort the result by total spending in descending order.

Expected Output Columns: Customer Name | City | Order Count | Total Spending

Concepts Tested: JOIN between Customers & Orders, GROUP BY, HAVING with COUNT and SUM, ORDER BY DESC

Write query and output in separate A4 sheets provided.

QUESTION 2: AI Model Monitoring (Zoho / Freshworks AI Team Pattern)

Business Context: Identify underperforming AI models that need retraining. Daily work for AI Engineers in Zoho / Freshworks to monitor deployed models.

Table: Model_Deployments	Columns: model_id, model_name, accuracy, latency_ms, deployed_city, last_trained_date, status
Sample Data	M01, BERT-Chat, 92.5, 120, Chennai, 2026-06-01, Active M02, YOLO-Vision, 65.0, 250, Bangalore, 2026-03-15, Active M03, XGBoost-Fraud, 58.0, 80, Chennai, 2026-01-10, Active M04, GPT-Finetune, 91.0, 300, Coimbatore, 2026-07-20, Active

Question: Find all Active models where accuracy is less than 70% OR latency is greater than 200ms. Categorize them using CASE statement as: - 'Critical - Retrain' if accuracy < 60 - 'Latency Issue' if latency > 200 - Else 'Monitor' Also calculate number of days since last trained using DATEDIFF.

Expected Output Columns: Model Name | Accuracy | Latency | Category | Days_Since_Trained

Concepts Tested: WHERE with OR, CASE WHEN, DATEDIFF function

Write query and output in separate A4 sheets provided.

QUESTION 3: HR Analytics - TCS iON Pattern (Second Highest Salary)

Business Context: Salary benchmarking and finding Nth highest salary - Most frequently asked in TCS NQT, Infosys SP, Wipro Elite. Tests subquery logic.

Table: Employees	emp_id, name, department, salary, experience, location
Sample Data	101, Arun, AI, 65000, 2, Chennai 102, Priya, Data, 72000, 3, Bangalore 105, Karthik, Data, 85000, 5, Bangalore 106, Divya, AI, 78000, 4, Chennai 103, Rahul, AI, 58000, 1, Coimbatore

Question: Part A (4 Marks): Find department-wise average salary where average salary is greater than ■60,000. Part B (6 Marks): Find the employee(s) with the SECOND highest salary in the AI department WITHOUT using LIMIT / TOP keyword. You must use subquery logic.

Expected Output Columns: Part A: Department | Average Salary Part B: Employee Name | Department | Salary

Concepts Tested: Part A - GROUP BY + HAVING AVG | Part B - Subquery with MAX < MAX

Write query and output in separate A4 sheets provided.

QUESTION 4: RPA Bot Failure Analysis (Accenture / UiPath Pattern)

Business Context: Bot downtime analysis using Window Functions. Core skill for AI & Automation domain - Predictive Maintenance of bots.

Table: Bot_Logs	log_id, bot_id, bot_name, tasks_completed, failed_tasks, execution_date
Sample Data	1, B01, InvoiceBot, 120, 5, 2026-08-20 2, B02, HRBot, 80, 12, 2026-08-20 3, B01, InvoiceBot, 150, 2, 2026-08-21 4, B03, DataScraper, 200, 20, 2026-08-20 5, B03, DataScraper, 210, 8, 2026-08-21 6, B04, SupportBot, 95, 0, 2026-08-20

Question: For each bot, show execution_date, failed_tasks, and previous day's failed_tasks using LAG() window function. Then filter and show only those logs where failed_tasks is greater than previous day's failed_tasks (failure increasing). Add a column Failure_Trend as 'Increasing' or 'Decreasing' using CASE.

Expected Output Columns: Bot Name | Execution Date | Failed Tasks | Previous Day Failed | Failure_Trend

Concepts Tested: CTE (WITH clause) + LAG() OVER(PARTITION BY bot_id ORDER BY execution_date) + CASE WHEN

Write query and output in separate A4 sheets provided.

QUESTION 5: Data Cleaning & Quality Check (Cognizant / Wipro Pattern)

Business Context: Real-world data is dirty. MNCs like Cognizant and Wipro test NULL handling and duplicate detection. Essential for AI/ML data preprocessing role.

Table: Customer_Feedback	feedback_id, customer_id, rating, feedback_text, city
Sample Data	1, C01, 5, 'Good Service', Chennai 2, C02, NULL, 'Nice', Bangalore (NULL rating) 3, C01, 4, 'Average', Chennai (Duplicate customer) 4, C03, 3, '', Coimbatore (Blank feedback) 5, C04, NULL, NULL, Chennai (Both NULL)

Question: Part A (5 Marks): Find number of valid feedbacks per city, but exclude records where rating is NULL and feedback_text is blank or NULL. Also write a separate query to count how many records have NULL ratings. Part B (5 Marks): Find customer_ids who have given more than 1 feedback (duplicate customers). Show their feedback count.

Expected Output Columns: Part A: City | Feedback Count and NULL Rating Count Part B: Customer ID | Feedback Count

Concepts Tested: IS NOT NULL, IS NULL, <> '', GROUP BY, HAVING COUNT(*) > 1

Write query and output in separate A4 sheets provided.

QUESTION 6: Sales Leadership Dashboard (Infosys / CTS Pattern - CTE + RANK)

Business Context: Find top performer per city and overall ranking. Leadership dashboard query frequently asked in Infosys and Cognizant to test CTE and RANK().

Table: Sales	sale_id, salesperson, city, product_category, amount, sale_date
Sample Data	1, Arun, Chennai, Electronics, 65000, 2026-08-01 2, Priya, Bangalore, Electronics, 35000, 2026-08-02 3, Karthik, Bangalore, Furniture, 80000, 2026-08-03 4, Divya, Chennai, Electronics, 90000, 2026-08-04 5, Arun, Chennai, Furniture, 40000, 2026-08-05 6, Priya, Bangalore, Furniture, 60000, 2026-08-06

Question: Using a CTE, first calculate total sales per salesperson per city. Then using RANK() window function, find the top 1 salesperson per city based on total sales. Also show overall rank of each salesperson across all cities.

Expected Output Columns: City | Salesperson | Total Sales | City Rank | Overall Rank

Concepts Tested: CTE (WITH SalesAgg AS...), GROUP BY city, salesperson, SUM(amount), RANK() OVER(PARTITION BY city ORDER BY SUM DESC), RANK() OVER(ORDER BY SUM DESC)

Write query and output in separate A4 sheets provided.

